Solar Suitability Zoning Map with Free Data Sources

1 Introduction

The global transition toward renewable energy has positioned solar power as a cornerstone of sustainable development strategies. For countries like Bangladesh, which faces rising energy demands, frequent power shortages, and a commitment to reducing carbon emissions, harnessing solar energy is not merely an alternative but a necessity. However, the efficiency and economic viability of solar photovoltaic (PV) installations are profoundly influenced by location-specific factors. Installing solar panels without a rigorous spatial analysis can lead to suboptimal energy generation, increased costs, and potential socio-environmental conflicts.

Geographic Information Systems (GIS) have emerged as a powerful tool for renewable energy planning, enabling the integration and analysis of multiple geospatial variables to identify optimal sites. Traditional approaches often rely on single factors like solar irradiance, but true suitability requires a holistic Multi-Criteria Decision Analysis (MCDA) that balances energy potential with practical constraints. These constraints include land availability, proximity to existing infrastructure for ease of connection and maintenance, and avoidance of environmental or agricultural sensitivities.

This study addresses this need by developing a comprehensive, high-resolution solar panel suitability zoning model for Sirajganj District, Bangladesh. It moves beyond basic solar mapping to create a nuanced decision-support tool that synthesizes physical potential (solar irradiance, PV output), land-use practicality, and infrastructural economics into a single, actionable map. The methodology explicitly details the treatment of both continuous raster data and discrete vector data, including the rationale for choosing Inverse Distance Weighting (IDW) over kriging for proximity-based criteria. Furthermore, the model is validated using spatial statistics (Optimized Hotspot Analysis), ensuring the robustness and reliability of the final suitability zones for end-users, including energy planners, private investors, and policymakers.

1.1 Objectives

The primary objective of this research is to delineate and classify zones within Sirajganj District based on their suitability for the installation of ground-mounted solar photovoltaic (PV) panels. This overarching goal is achieved through the following specific objectives:

1. To collect, preprocess, and standardize key geospatial datasets, including Global Horizontal Irradiance (GHI), Photovoltaic Output Potential (PV), Land Use/Land Cover (LULC), road networks, and residential buildings.
2. To develop a standardized scoring and reclassification scheme (1-5) for each criterion, accurately reflecting its influence on solar farm suitability.
3. To implement a GIS-based Weighted Overlay Analysis, integrating the reclassified criteria with scientifically justified weights (GHI: 40%, LULC: 25%, PV Output: 15%, Proximity to Roads: 10%, Proximity to Houses: 10%) to generate a composite suitability index.
4. To validate the results of the MCDA model using Optimized Hotspot Analysis on core input variables (GHI and suitable LULC classes) to confirm the statistical significance of identified high-suitability zones.

2 Methodology

The methodology follows a structured multi-criteria GIS analysis, combining both direct measurement variables and derived proximity criteria. The overall workflow is illustrated in Figure 1 and detailed in the following subsections.

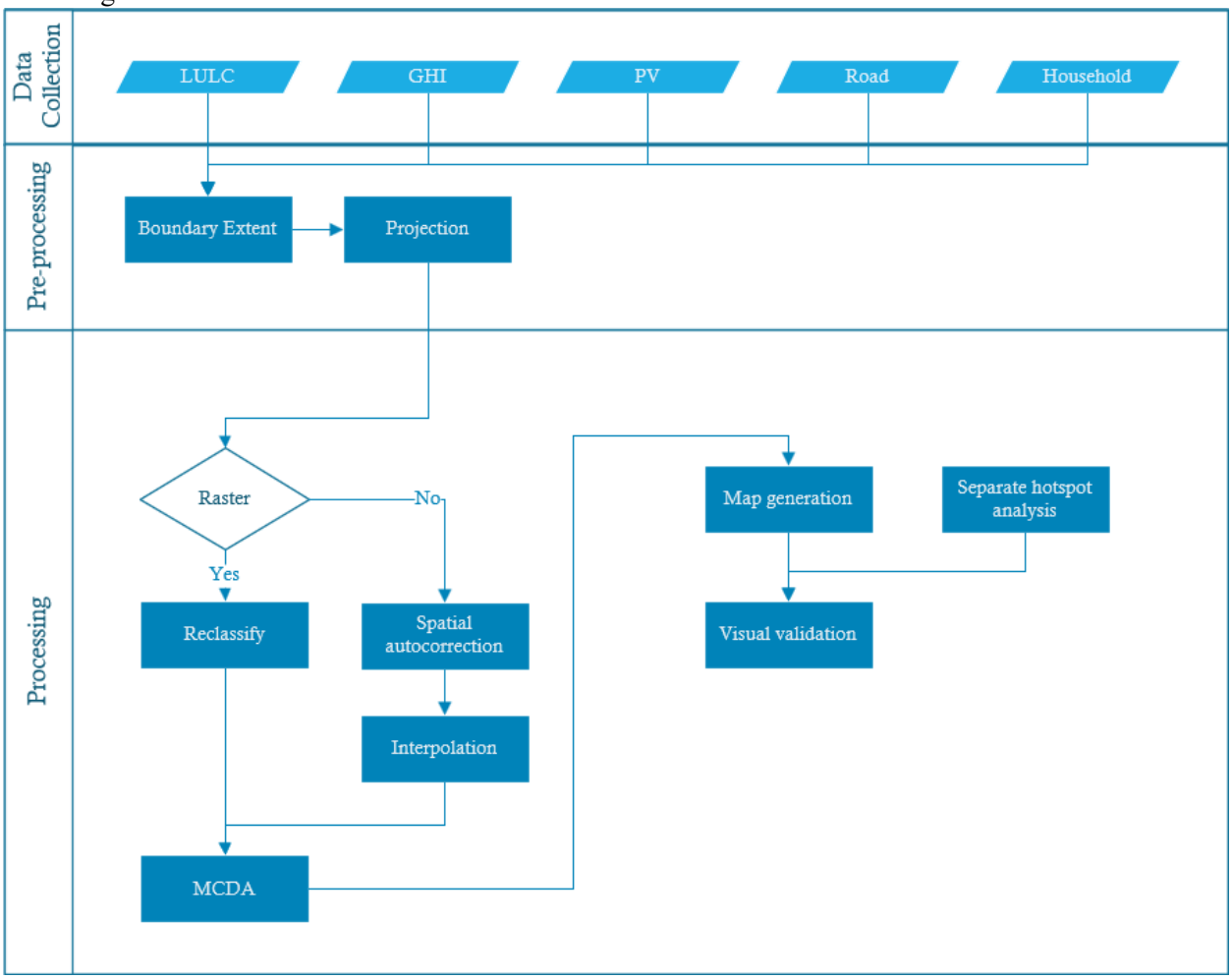


Figure 1: Methodological Framework

2.1 Data Acquisition and Preprocessing

Five primary datasets were acquired:

- Global Horizontal Irradiance (GHI) Raster: Annual average solar energy received (kWh/m²/day). Source: Global Solar Atlas.
- Photovoltaic Output Potential (PV) Raster: Simulated annual PV power output (kWh/kWp). Source: Global Solar Atlas.
- Land Use/Land Cover (LULC) Raster: Categorized classes (e.g., barren land, grassland, built-up, forest, water). Source: Sentinel 2 LULC
- Road Network (Vector): Line features of all transport pathways. Source: OpenStreetMap
- House/Building Footprints (Vector): Polygon features of residential/commercial structures. Source: OpenStreetMap

All datasets were projected to a common coordinate reference system (WGS 84 / UTM zone 45N).

2.2 Criterion Standardization and Reclassification

Each criterion layer was standardized into a common suitability scale of 1 (Least Suitable) to 5 (Most Suitable).

- For Direct Raster Criteria (GHI, PV, LULC):
  These gridded datasets represent continuously measured phenomena. Since they were already in a spatially exhaustive raster format with values at every location, spatial autocorrelation was inherently accounted for in their original generation. Modeling spatial autocorrelation (e.g., via Kriging) is only necessary if one is interpolating from sparse point samples to create a continuous surface. As this was not required here, the rasters were directly reclassified using defined thresholds.
  - GHI & PV: Higher values indicate higher suitability. Quantile or natural break classification was used to divide values into 5 classes.
  - LULC: A suitability score was assigned per class based on energy policy, environmental, and economic factors (e.g., Barren Land: 5, Grassland: 4, Agricultural Land: 3, Built-up: 2, Forest/Waterbody: 1).
- For Distance-Based Vector Criteria (Roads, Houses):
  These vector layers represent discrete features. Suitability is not inherent to their location but is a function of proximity. Therefore, they require transformation.
  - Euclidean Distance Raster Creation: Distance rasters were generated, where each pixel's value represents its straight-line distance to the nearest road or house.
  - Interpolation Consideration: The distance surface is a deterministic function, not a spatially autocorrelated statistical phenomenon. Using Kriging would be inappropriate as it models spatial continuity and variance based on measured sample points, which does not apply to a calculated distance field. Instead, Inverse Distance Weighting (IDW) is a more suitable deterministic method for converting vector points (like distance centroids) to a raster, as it explicitly uses distance in its weighting function, aligning with the proximity-based logic of the criterion.
  - Reclassification: For roads, closer proximity is more suitable (for ease of access and grid connection). Same for the household. Reclassification was done to match with the 5 class criterion.

2.3 Spatial Autocorrelation Consideration

Had the GHI, PV, or LULC data been available only as point samples, spatial interpolation would have been necessary. In such a scenario, examining spatial autocorrelation via semivariogram analysis is a critical first step. If spatial autocorrelation is present (i.e., nearer points are more similar), Ordinary Kriging would be the optimal geostatistical method to create a predictive surface, as it provides a best linear unbiased estimate and a measure of prediction error (kriging variance). If no spatial autocorrelation is detected, other methods like IDW or even simple averaging would be considered. This study's use of pre-existing, full-coverage rasters bypassed this need but the conceptual step is vital for methodological rigor.

2.4 Multi-Criteria Decision Analysis (Weighted Overlay)

The reclassified raster layers were integrated using the Weighted Overlay tool in GIS. The weight assigned to each criterion reflects its relative importance in determining overall suitability, derived from expert judgment or analytical methods like AHP.

Weight Assignment:

- GHI (40%): Highest weight, as it is the fundamental physical driver of solar energy yield.
- LULC (25%): Critical constraint layer, determining land availability and environmental/social acceptability.
- PV Output (15%): Reflects technology-specific efficiency influenced by local climate (temperature, etc.).

- Proximity to Roads (10%): Influences installation and maintenance cost.
- Proximity to Houses (10%): Influences potential for distributed generation and energy loss in transmission.

2.5 Validation using Optimized Hotspot Analysis

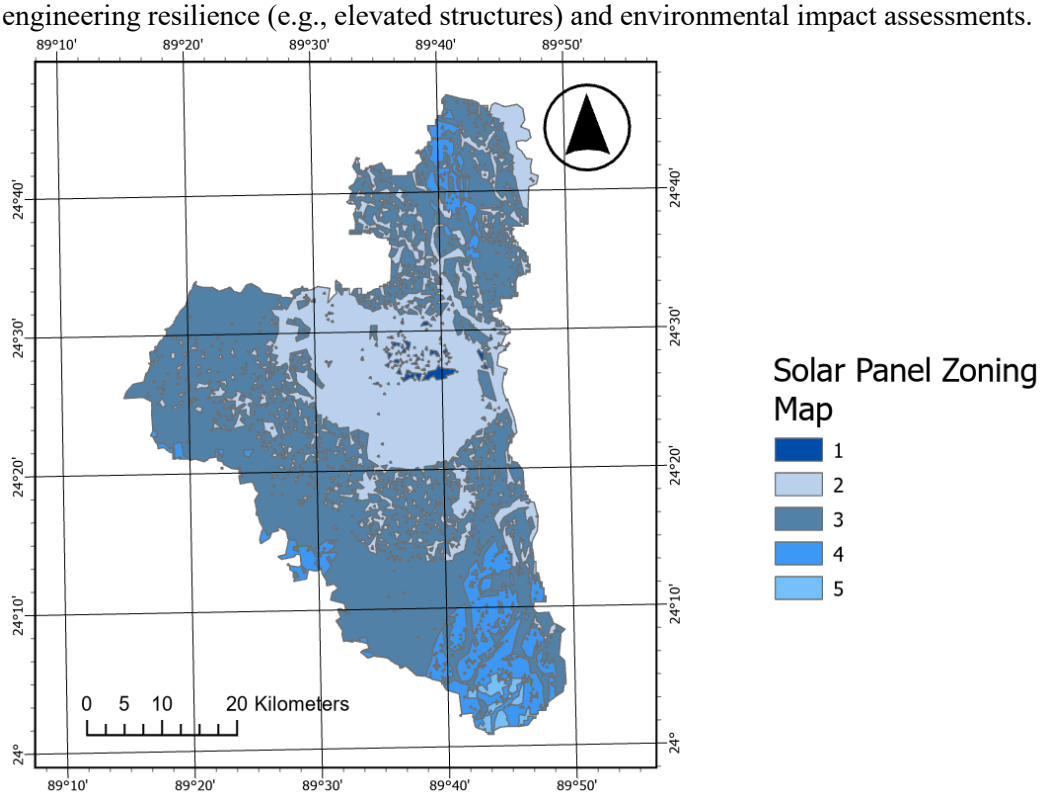
To cross-validate the results, Optimized Hotspot Analysis (Getis-Ord Gi*) was performed on two key input layers: GHI and LULC (using their original values). This technique identifies statistically significant spatial clusters of high values (hotspots) and low values (coldspots) while automatically correcting for spatial scale and distributional issues.

- The GHI hotspot map confirmed areas of consistently high solar radiation.
- The LULC hotspot map (for suitable classes) identified clusters of available land.

A visual and statistical comparison showed strong spatial concordance between the composite suitability map's "Most Suitable" zones and the overlapping hotspots of high GHI and suitable LULC, lending credibility to the model's output.

3 Results & Discussion

The results present a realistic and actionable plan for solar development in Sirajganj. The prominence of river chars as high-suitability zones is a key finding. While these lands are often underutilized and have excellent solar potential, their ecological sensitivity and vulnerability to flooding must be addressed through engineering resilience (e.g., elevated structures) and environmental impact assessments.



The model's strength lies in its transparent, replicable methodology. The explicit weighting (40% GHI) correctly prioritizes fundamental energy yield, while the significant LULC weight (25%) embeds a land-conservation ethic. The use of IDW for proximity rasters was appropriate, as it directly models the decreasing influence of a feature with distance, which aligns with the economic and logistical logic of the "proximity to roads" criterion.

The model uses static datasets; incorporating temporal flooding models would refine suitability on the chars. Furthermore, the weights, though justified, were based on expert judgment. Future work could employ the Analytical Hierarchy Process (AHP) with a broader panel of stakeholders to derive consensus-based weights. Finally, integrating detailed grid connectivity data would enhance the practical utility of the map for utilities.

This study successfully identifies clear, prioritized zones for solar investment. It recommends that policymakers and investors first target the identified Class 5 zones on stable river chars for large-scale projects, initiate pilot agrivoltaic projects in Class 3 agricultural zones, and entirely avoid the Class 1 and 2 zones to prevent resource conflict and ensure project viability. This zoning map serves as a vital scientific foundation for Sirajganj's sustainable energy transition.

Figure 2: Sirajganj solar panel zoning map

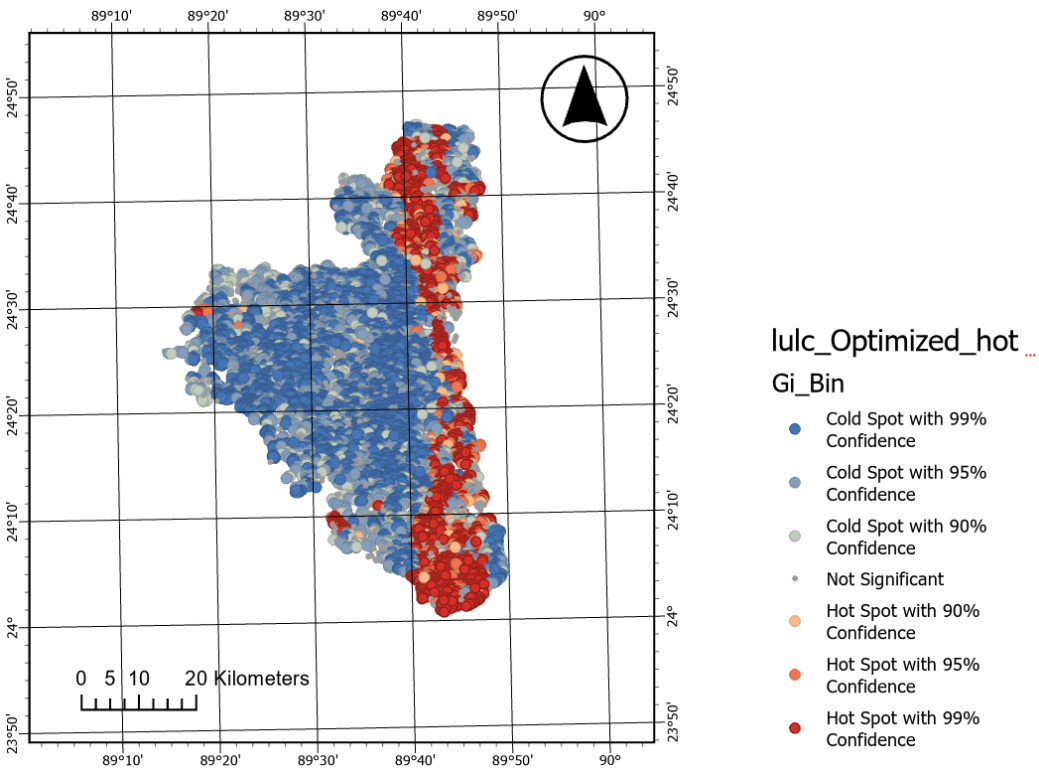


Figure 3: Optimized Hotspot analysis with land cover data

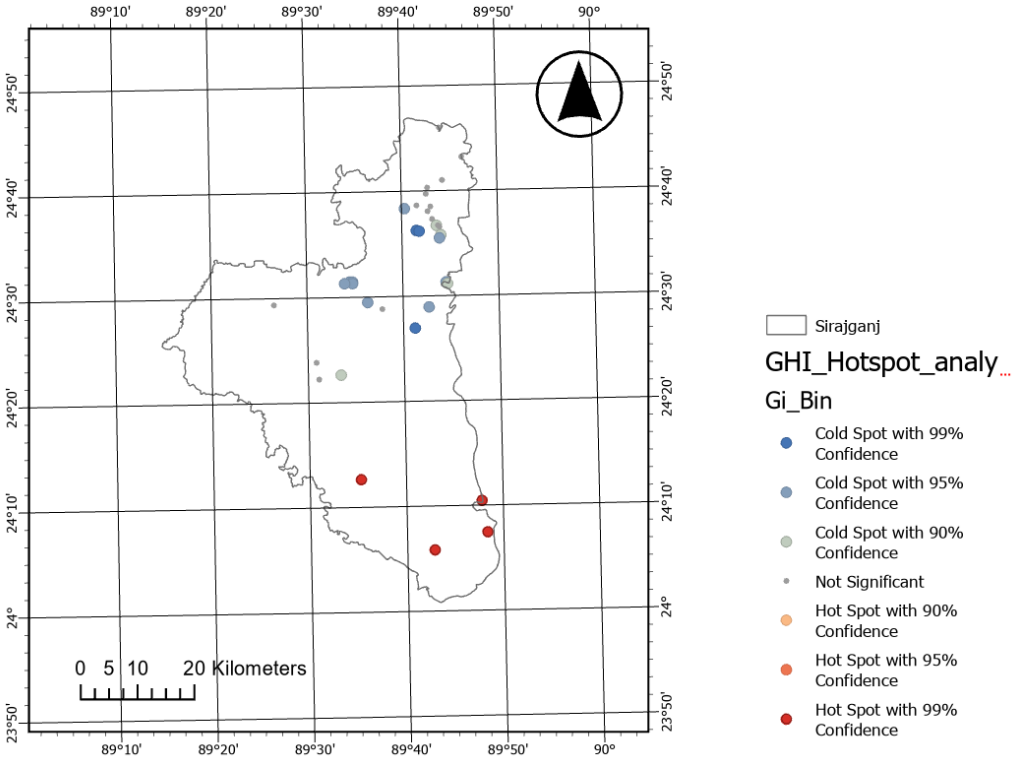


Figure 4:Hotspot analysis with GHI